

# FUNCTIONAL\_SPECIFICATION

## Functional Specification

Status legend: **Implemented** · **Partial** · **Stub** / **Planned**

### 1. Authentication & session

| Capability                             | Status      | Notes                      |
|----------------------------------------|-------------|----------------------------|
| Staff login (username/password)        | Implemented | JWT returned               |
| Current user profile (/auth/me)        | Implemented |                            |
| Change password                        | Implemented | Min 8 chars; confirm match |
| Remember-me (local vs session storage) | Implemented | Frontend                   |
| Failed-attempt logout                  | Implemented | LoginAttemptService        |
| Role-based route guards                | Implemented | Angular authGuard          |

### 2. Dashboard

| Capability                                      | Status      | Notes                                       |
|-------------------------------------------------|-------------|---------------------------------------------|
| Summary counts (customers, accounts, employees) | Implemented |                                             |
| Today's transaction count                       | Partial     | Hardcoded 0 until Transaction module exists |

### 3. Customers

| Capability                           | Status      | Notes                                                       |
|--------------------------------------|-------------|-------------------------------------------------------------|
| Search / paginate customers          | Implemented |                                                             |
| View customer detail                 | Implemented |                                                             |
| Create customer                      | Implemented | Optional first account on create                            |
| Update customer                      | Implemented |                                                             |
| Soft/hard delete                     | Implemented | ADMIN only for DELETE                                       |
| Open additional accounts from detail | Implemented | Current (policy-aware opening deposit); Loan button present |

### 4. Accounts

| Capability             | Status      | Notes                            |
|------------------------|-------------|----------------------------------|
| Open account           | Implemented | Policy-validated opening deposit |
| List / search accounts | Implemented | /app/accounts                    |

| Capability                             | Status      | Notes                                |
|----------------------------------------|-------------|--------------------------------------|
| List accounts by customer              | Implemented |                                      |
| Update account status                  | Implemented | API present                          |
| Deposit (post-open)                    | Implemented | Credits balances only; no ledger row |
| Withdraw / transfer                    | Stub        | Not built                            |
| Account detail + ledger UI             | Stub        | Not built                            |
| Account policies (create + get active) | Implemented | Seeded INR policies; no LOAN seed    |
| Opening-deposit dialog (Current)       | Implemented | Uses GET /account-policies           |

## 5. Employees (users)

| Capability                       | Status      | Notes                                                   |
|----------------------------------|-------------|---------------------------------------------------------|
| List / create / update employees | Implemented | ADMIN                                                   |
| Role assignment                  | Implemented | Seeded roles include unused TELLER / AUDITOR / CUSTOMER |

## 6. Branch

| Capability         | Status | Notes                               |
|--------------------|--------|-------------------------------------|
| Branch master data | Stub   | Only branchCode string (often 0001) |

## 7. Loans

| Capability                       | Status  | Notes                                                 |
|----------------------------------|---------|-------------------------------------------------------|
| Loan product / EMI / collateral  | Stub    |                                                       |
| AccountType.LOAN account opening | Partial | Enum + UI affordance; needs policy; not a loan domain |

## 8. Transactions & reports

| Capability                    | Status | Notes                                     |
|-------------------------------|--------|-------------------------------------------|
| Transaction ledger entity/API | Stub   | Empty transaction package                 |
| Reports                       | Stub   | Empty report package; sidebar placeholder |
| Beneficiaries                 | Stub   | Empty beneficiary package                 |

## 9. Audit

| Capability                    | Status  | Notes                            |
|-------------------------------|---------|----------------------------------|
| JPA created/updated by fields | Partial | AuditableEntity on some entities |

|  | Capability           | Status | Notes               |
|--|----------------------|--------|---------------------|
|  | Audit event API / UI | Stub   | Empty audit package |

## Non-functional (current)

| Area    | Behavior                                                                       |
|---------|--------------------------------------------------------------------------------|
| AuthN   | JWT HMAC, 1h expiry ( <code>jwt.expiration</code> )                            |
| AuthZ   | Role matchers in <code>SecurityConfig</code>                                   |
| CORS    | <code>localhost</code> , <code>bankone.local</code> , <code>192.168.0.4</code> |
| Logging | Hibernate SQL DEBUG (dev-oriented)                                             |
| Deploy  | Open Liberty WAR primary path                                                  |

Deferred hardening items are tracked in `.cursor/rules/deferred-hardening.mdc`.